Introduction to Operations Research, Deterministic Models

Lecture 3: Simplex Method

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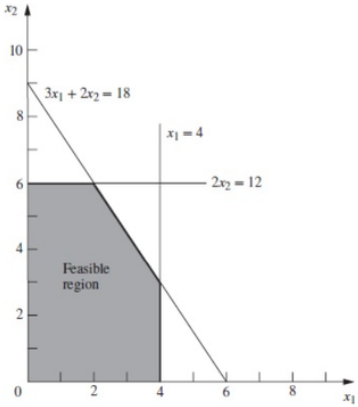
Recap: geometric concepts



For an LP problem with n decision variables,

- each **constraint boundary** is a line that forms the boundary of what is permitted by the corresponding constraint.
- each of its **corner-point solutions** lies at the intersection of n constraint boundaries. some of which within the feasible region are **corner-point feasible** (CPF) solutions, and others are corner-point infeasible solutions.
- two CPF solutions are **adjacent** to each other if they share $n-1$ constraint boundaries. The two adjacent CPF solutions are connected by a line segment that lies on these same shared constraint boundaries. Such a line segment is referred to as an **edge** of the feasible region.

Optimality test: For an LP problem with at least one optimal solution, if a CPF solution has no adjacent CPF solutions that are better, then it must be an optimal solution.

Recap: geometric outline

Initialization: choose $(0, 0)$ as the initial CPF solution to examine.

Optimality test: find the two CPF solutions (0,6) and (4,0) that are adjacent to $(0, 0)$;

1. Choose the edge to move along with
2. Stop at the first new constraint boundary
3. Compute the intersection of the new set of constraint boundaries

Both two adjacent CPF solutions (0,6) and (4,0) obtain better objective values, thus (0,0) is not an optimal solution.

Iteration 1: move to a better adjacent CPF solution, say (0,6).

Optimality test: the CPF solution (2,6) that adjacent to (0,6) obtains better objective value, thus (0,6) is not an optimal solution.

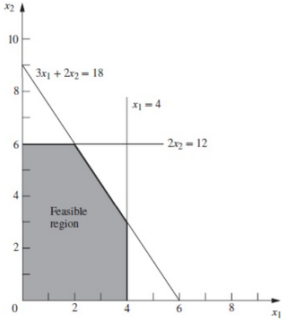
Iteration 2: move to a better adjacent CPF solution (2,6).

Optimality test: None of adjacent CPF of (2,6) achieves the better objective value, thus we conclude that (2,6) is an optimal solution.

Recap: Algebraic concepts

Augmented form of the Wyndor model

$$
\begin{array}{ll}
\text{maximize} & Z = 3x_1 + 5x_2 \\
\text{subject to} & x_1 \qquad\qquad + x_3 = 4 \\
& \qquad 2x_2 \qquad + x_4 = 12 \\
& 3x_1 + 2x_2 \qquad\qquad + x_5 = 18 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5.
\end{array}
$$



- x_3, x_4, x_5 are referred to as *slack variables* which are used to transform $\leq$ inequality into equation with non-negativity constraints.

- An **augmented solution** is a solution for the original variables that has been augmented by the corresponding values of the slack variables. e.g. (3, 2, 1, 8, 5)

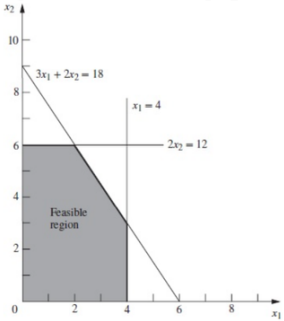
- a **basic solution** is an augmented corner-point solution; e.g. (4, 6, 0, 0, -6)

- a **basic feasible (BF) solution** is an augmented corner-point feasible (CPF) solution; e.g.

Simplex method: algebraic concepts

Basic and nonbasic variables

$$
\begin{array}{ll}
\text{maximize} & Z = 3x_1 + 5x_2 \\
\text{subject to} & x_1 \qquad\quad + x_3 = 4 \\
& \qquad 2x_2 \quad + x_4 = 12 \\
& 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5.
\end{array}
$$



- For a general LP problem with n decision variables and m functional inequality constraints,
  - # of variables in the augmented form: $m + n$
    # of constraint boundaries: $m + n$
  - each corner-point solution lies at the intersection of n constraint boundaries; and the corresponding n (slack or original) variables take value 0;
  - for each basic (augmented corner-point) solution, the corresponding n variables that takes value 0 are called nonbasic variables;
  - the other m variables in the augmented model are called basic variables, whose value can be obtained by solving the m equations.

Basic and nonbasic variables

Properties:

• Each variable is designated as either a nonbasic variable, or a basic variable.

• The number of nonbasic variables equals n (the number of variables in the general form). The value of nonbasic variables are set equal to 0.

• The number of basic variables is m (the number of functional constraints in the general form). The value of basic variables are obtained as the simultaneous solution of the system of equations.

• If the basic solution satisfies the nonnegativity constraints, the basic solution is a basic feasible (BF) solution.

Question: Can we arbitrarily choose a set of nonbasic and basic variables?

Example: $(0, 6, 4, 0, 6)$

- $(0, 6)$ is a CPF solution, and $(0, 6, 4, 0, 6)$ is an augmented CPF solution.
- Let x_1 and x_4 be two nonbasic variables with value 0, then the three equations

$$x_3 = 4, \quad 2x_2 = 12, \quad 2x_2 + x_5 = 18$$

yields the solution for the three basic variables, $x_3 = 4$, $x_2 = 6$, $x_5 = 6$.

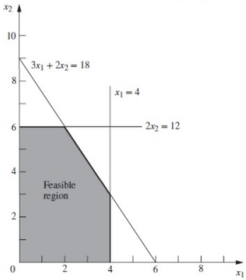
Adjacent basic variables

- Two CPF solutions are **adjacent** to each other if they share $n-1$ constraint boundaries.
- Two BF solutions are **adjacent** if they share <u>all but one</u> nonbasic variables, and consequently, they share <u>all but one</u> basic variables.

$$
\begin{array}{ll}
\text{maximize} & Z = 3x_1 + 5x_2 \\
\text{subject to} & x_1 \qquad\quad + x_3 = 4 \\
& \qquad 2x_2 \quad + x_4 = 12 \\
& 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5.
\end{array}
$$

Example:

- $(0, 6)$ and $(0, 0)$ are two adjacent CPF solutions;
- $(0, 6, 4, 0, 6)$ is an augmented solution with nonbasic variables x_1 and x_4.
- $(0, 0, 4, 12, 18)$ is another augmented solution with nonbasic variables x_1 and x_2.
- $(0, 6, 4, 0, 6)$ and $(0, 0, 4, 12, 18)$ share one common nonbasic variables, and thus they are adjacent BF solutions.



Algebraic procedure of the Simplex method

Conceptual outline of the Simplex Method

1. **Initialization** to identify the initial solution for starting the simplex method.

For iteration $k = 0, 1, 2, \ldots,$

2. **Apply the optimality test** to determine if the current solution is optimal.
   a. If so, stop
   b. If not, continue
3. **Determine which direction** to move to get to the next solution
4. **Determine where to stop** to reach this next solution
5. **Solve for this new solution**

Algebraic procedure of the Simplex method

$$
\begin{array}{ll}
\text{maximize} & Z = 3x_1 + 5x_2 \\
\text{subject to} & x_1 \qquad\quad + x_3 = 4 \\
& \qquad 2x_2 \quad + x_4 = 12 \\
& 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5.
\end{array}
$$

1. Initialization:

• Based on "solution concept 3", the initialization of the simplex method chooses the origin to be the initial CPF solution whenever possible.

• Let x_1 and x_2 be the two nonbasic variables, then we obtain an initial BF solution $(0, 0, 4, 12, 18)$.

Algebraic procedure of the Simplex method

$$
\begin{array}{ll}
\text{maximize} & Z = 3x_1 + 5x_2 \\
\text{subject to} & x_1 \qquad\quad + x_3 = 4 \\
& \qquad 2x_2 \quad + x_4 = 12 \\
& 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5.
\end{array}
$$

2. Apply the optimality test:

• If there is no positive rate of improvements of objective function value towards an adjacent BF(CPF) solution, then the current BF solution is optimal.

• Moving from a BF(CPF) solution towards adjacent BF(CPF) solution corresponds to increase the value of <u>one nonbasic variable</u> from 0 while the other nonbasic variables remain the same, and the values of basic variables are adjusted accordingly to satisfy the equation.

• At $(0, 0, 4, 12, 18)$, if we increase the value of either x_1 or x_2, the rate of improvements of objective function value $Z = 3x_1 + 5x_2$ are both positive. Thus $(0, 0, 4, 12, 18)$ is not optimal

Algebraic procedure of the Simplex method

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad\quad + x_3 = 4 \\ & \qquad 2x_2 \quad + x_4 = 12 \\ & 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{array}$$

3. Determine which direction to move:

• Based on "solution concept 6", the Simplex method chooses to move towards an adjacent BF solution with the largest positive rate of improvement in Z.

• Moving from the current BF solution along an edge towards an adjacent BF solution corresponds to increase the value of <u>one nonbasic variable</u> (it will be converted into a basic variable for the next adjacent BF solution, and thus is called the **entering basic variable**).

• At $(0, 0, 4, 12, 18)$, the rate of improvements for increasing x_1 or x_2 are 3 and 5. Thus we select x_2 as the entering basic variable.

Algebraic procedure of the Simplex method

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad\quad + x_3 = 4 \\ & \qquad 2x_2 \quad + x_4 = 12 \\ & 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{array}$$

4. Determine where to stop:

• Since the rate of improvement is positive, it determines the greatest value to increase the entering basic variable x_2 before leaving the feasible region. With $x_1 = 0$,

$$\begin{cases} x_1 \qquad\quad + x_3 = 4 \\ \qquad 2x_2 \quad + x_4 = 12 \\ 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\ x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{cases} \iff \begin{cases} x_3 = 4 \\ x_4 = 12 - 2x_2 \quad \Longrightarrow \quad x_2 \leq 12/2 = 6 \\ x_5 = 18 - 2x_2 \quad \Longrightarrow \quad x_2 \leq 18/2 = 9 \\ x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{cases}$$

To satisfy nonnegativity requirements, x_2 can only be increased to 6, and thus x_4 dropped to 0.

• Asumming the RHS are nonnegative, this computation process is referred to as **minimum ratio test**, which computes the ratio of the RHS to the strictly positive coefficient of the entering basic variable, and the basic variable with minimum ratio drops to 0 (which becomes a nonbasic variable for the next BF solution. and thus is called the **leaving basic variable**).

Algebraic procedure of the Simplex method

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad\quad + x_3 = 4 \\ & \qquad 2x_2 \quad + x_4 = 12 \\ & 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{array}$$

5. Solving for the new BF solution:

	Initial BF solution	New BF solution
Nonbasic variables	$x_1 = 0, x_2 = 0$	$x_1 = 0, x_4 = 0$
Basic variables	$x_3 = 4, x_4 = 12, x_5 = 18$	$x_2 = 6, x_3 = ?, x_5 = ?$

We need to perform some elementary algebraic operations to produce a new equivalent equation system such that

• the objective is expressed in terms of new nonbasic variables x_1 and x_4, so that in the next iteration it is easy to compare the rate of improvements

• each equation contains just one basic variable with the coefficient 1, so that it will immediately yield the new BF solution $(0, 6, 4, 0, 6)$ with $Z = 30 + 3x_1 - \frac{5}{2}x_4 = 30$

The Simplex method in tabular form

Rewrite the augmented form

Augmented form of the model

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad\quad + x_3 = 4 \\ & \qquad 2x_2 \quad + x_4 = 12 \\ & 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{array}$$

Assume the RHS coefficients are nonnegative.

It is equivalent to

$$\begin{array}{ll} \text{maximize} & Z \\ \text{subject to} & Z - 3x_1 - 5x_2 = 0 \\ & x_1 \qquad\quad + x_3 = 4 \\ & \qquad 2x_2 \quad + x_4 = 12 \\ & 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{array}$$

Simplex method in tabular form

$$
\begin{aligned}
\text{maximize} \quad & Z \\
\text{subject to} \quad & Z - 3x_1 - 5x_2 = 0 \\
& x_1 \qquad\quad + x_3 = 4 \\
& \qquad 2x_2 \quad + x_4 = 12 \\
& 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5.
\end{aligned}
$$

Iteration	Basic variable	row	Coefficient						RHS	ratio
			Z	x_1	x_2	x_3	x_4	x_5		
0	Z	(0)	1	-3	-5	0	0	0	0	
	x_3	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	3	2	0	0	1	18	

Main steps of the Simplex method in tabular form

1. **Initialization.** The initial BF solution is $(0, 0, 4, 12, 18)$

2. **Optimality test:** The current BF solution is optimal if and only if every coefficient in row 0 is nonnegative.

3. **Determine the entering basic variable** that has the "most negative" coefficient in row 0. The column below that coefficient is called pivot column.

4. **Determine the leaving basic variable** by applying the minimum ratio test as follows.

(a) pick out each coefficient in the pivot column that is strictly positive

(b) divide each of these coefficients into the RHS entry for the same row

(c) identify the row that has the smallest of these ratios, and the basic variable for that row is the leaving basic variable. (That row is called pivot row and the number that is in both pivot row and pivot column is called the pivot number.)

Main steps of the Simplex method in tabular form

5. **Solve for the new BF solution** using elementary row operations.

(a) Divide the pivot row by the pivot number. Use this new pivot row in the follow-up steps.

(b) For each other row (including row 0) that has a negative coefficient in the pivot column, add to this row the product of the absolute value of this coefficient and the new pivot row.

(c) For each other row that has a positive coefficient in the pivot column, subtract from this row the product of this coefficient and the new pivot row.

Simplex method in tabular form

Iteration	Basic variable	row	Coefficient						RHS	ratio
			Z	x_1	x_2	x_3	x_4	x_5		
0	Z	(0)	1	-3	-5	0	0	0	0	
	x_3	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	3	2	0	0	1	18	
1	Z	(0)	1	-3	0	0	5/2	0	30	
	x_3	(1)	0	1	0	1	0	0	4	
	x_2	(2)	0	0	1	0	1/2	0	6	
	x_5	(3)	0	3	0	0	-1	1	6	
2	Z	(0)	1	0	0	0	3/2	1	36	
	x_3	(1)	0	0	0	1	1/3	-1/3	2	
	x_2	(2)	0	0	1	0	1/2	0	6	
	x_1	(3)	0	1	0	0	-1/3	1/3	2	

Exercise: Simplex method

Use the simplex method to solve the following problem

$$\begin{array}{ll} \text{maximize} & Z = 2x_1 - x_2 + x_3 \\ \text{s.t.} & 3x_1 + x_2 + x_3 \leq 60 \\ & x_1 - x_2 + 2x_3 \leq 10 \\ & x_1 + x_2 - x_3 \leq 20 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

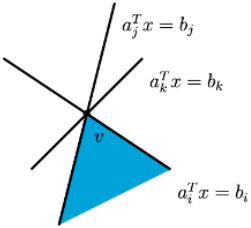
Tie breaking in Simplex

- Tie for entering basic variables: It is possible that two or more nonbasic variables are tied with the same largest negative coefficient. Then we select one of them arbitrarily.

- Tie for leaving basic variables: It is possible that two or more basic variables are tied with the same smallest ratio, and this case is referred to as degeneracy. It may result in a perpetual loop theoretically, but it rarely occurs in practice.

  - **Degeneracy in geometry:** the degenerate solution satisfies two different sets of equality constraints (lies on the intersection of more than n constraints)

  - **Bland's rule:** (a) The entering variable should be the lowest index variable with positive reduced cost. (b).The leaving variable (in case of a tie in the min ratio test) should be the lowest index row.



Special cases: no leaving basic variable

In the step of **determining the leaving basic variable**, it is possible that the entering basic variable could be increased to infinity without violating any functional or nonnegativity constraints.

This could happen when every coefficient in the pivot column is either negative or zero.

Iteration	Basic variable	row	Coefficient						RHS	ratio
			Z	x_1	x_2	x_3	x_4	x_5		
0	Z	(0)	1	-3	-5	0	0	0	0	
	x_3	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	-2	0	1	0	12	
	x_5	(3)	0	3	-2	0	0	1	18	

With the entering basic variable being increased to infinity, the objective value is unbounded.

The real message for practical problems is that a mistake has been made, such as omitting relevant constraints or stating them incorrectly.

Special cases: multiple optimal solution

- Any linear programming problem with multiple optimal solutions and a bounded feasible region has at least two CPF solutions that are optimal.
- Every optimal solution lies on the line segment connecting the two optimal CPF solutions.
- Whenever a problem has more than one optimal BF solution, at least one of the nonbasic variables has a zero coefficient in the final row 0, so increasing any such variable will not change the value of Z. So other optimal BF solutions can be identified by choosing a nonbasic variable with a zero coefficient as the entering basic variable.

Iteration	Basic variable	row	Coefficient						RHS	ratio
			Z	x_1	x_2	x_3	x_4	x_5		
2	Z	(0)	1	0	0	0	0	1	36	
	x_3	(1)	0	0	0	1	1/3	-1/2	2	
	x_2	(2)	0	0	1	0	1/2	0	6	
	x_1	(3)	0	1	0	0	-1/3	1/3	2	

Simplex methods for non-general LP models

Simplex methods for non-general LP models

The standard simplex method is performed for LP model with the following features:

1. The objective function is to be maximized
2. The functional constraints are in $\leq$ form with nonnegative right-hand sides.
3. The variables have nonnegativity constraints.

What if some of these features is not satisfied?

• The minimization $Z = \sum_{j=1}^{n} c_j x_j$ can be transformed into maximizing $Z' = \sum_{j=1}^{n} (-c_j) x_j$.

• The serious problem resulted from the absence of other features lies in identifying an initial BF solution.

• **Artificial-variable technique** is used to enable obtaining an initial BF solution so that the simplex method can proceed.

Idea of Artificial-variable technique

1. **Introduce an artificial variable** into each functional constraint that does not meet the required feature, which enlarges the feasible region unless the artificial variable is equal to 0. This leads to a revised LP problem and it is then easy to identify an initial BF solution for the revised problem.

2. **Drive the values of the artificial variables to 0** by using special methods (big-M method and two-phase method) to enable the simplex method to convert the revised problem back into the original problem, and identify a BF solution for the original problem.

3. The simplex method can proceed with that BF solution to solve the original problem.

Reading materials

Section 4.1, 4.2, 4.3, 4.4, *Introduction to Operations Research* by Hillier and Lieberman